UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. Which declaration creates an integer array of five elements?
   A. int a(5);
   B. int a[5];
   C. int a;
   D. array a[5];

2. Which data type stores a whole number?
   A. float
   B. char
   C. string
   D. int

3. Which operator reads data from cin?
   A. =
   B. >>
   C. %
   D. <<

4. Which part of a for loop is executed after each iteration?
   A. The update expression
   B. The condition only
   C. The initialisation
   D. The loop body twice

5. Which declaration creates a floating-point variable named price?
   A. char price;
   B. float price;
   C. bool price;
   D. int price;

6. What does a compiler do?
   A. Stores only integer values
   B. Draws flowcharts automatically
   C. Translates the whole source program before execution
   D. Executes one source statement at a time

7. Which values can a bool variable hold?
   A. any character
   B. true or false
   C. any integer
   D. any decimal

8. Which phase of the program development life cycle defines what the program must do?
   A. Testing
   B. Coding
   C. Maintenance
   D. Analysis

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9. Which structure chooses between two alternative paths?
   A. for
   B. array
   C. if...else
   D. struct

10. How many times does a while loop execute when its condition is false initially?
   A. Until the user stops it
   B. Zero
   C. Two
   D. One

PART B (60 MARKS)

QUESTION 1

This question tests high-level languages and debugging.

a) Define syntax error and logic error.

(3 marks)

b) Explain TWO advantages of high-level languages over machine language.

(2 marks)

QUESTION 2

A program reads a positive integer N and calculates the sum and average of the numbers 1 to N.

a) Draw a flowchart for the sum and average calculation.

(6 marks)

b) Write pseudocode and a complete C++ program for the algorithm.

(7 marks)

QUESTION 3

This question tests type conversion and constants.

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

(6 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

(5 marks)

QUESTION 4

Study the following C++ program segment:

```
int total = 0;
for (int i = 1; i <= 5; i++) {
   total += i;
}
cout << total;
```

a) Trace the program and state the output.

(7 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

(8 marks)

QUESTION 5

A sports club stores sport names and medal counts in parallel arrays.

a) Declare two parallel arrays for five sports and initialise them with suitable values.

(8 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(8 marks)

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PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER